7. a) If $f: R \to R$ is given by $f(x) = 3x - 5$, find a formula for

$f^{-1}(x) =$ $(x+5)/3$ (if it exists).

(6)

b) If $f: R \to R$ is given by $f(x) = x^2 - 4$, find a formula for

$f^{-1}(x) =$ dne (if it exists).

since $D(f^{-1}) = [-4, \infty)$ (niether one-to-one nor onto)

8. Below are the steps in the proof of a statement about lattices. Justify each step in terms of the definitions and basic properties of lattices

(12)

$a*b \leq b$ by definition of GLB(a,b)

$b \leq b+d$ by def^{n} of LUB$\{b,d\}$

$a*b \leq b+d$ by transitivity of $\leq$

Similarly, $a*b \leq a+c$ (no reason need be given)

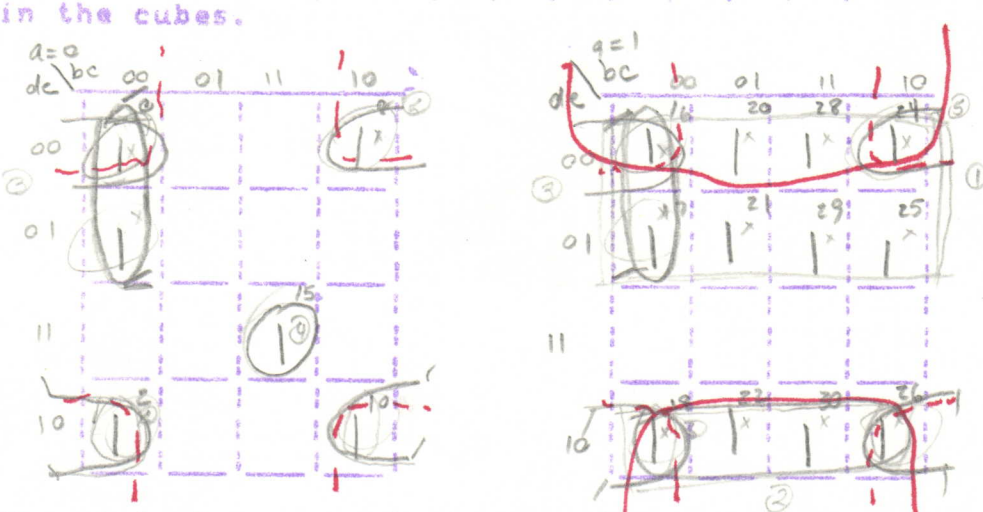
$a*b \leq (b+d)*(a+c)$ a*b is LB of (b+d) and (a+c). so by def^{n} of GLB, it is "below" the GLB$\{(a+c), (b+d)\}$

Similarly, $c*d \leq (b+d)*(a+c)$ (no reason need be given)

$(a*b)+(c*d) \leq (b+d)*(a+c)$ (b+d)*(a+c) is UB of (a*b) and (c*d) so by def^{n} of LUB it is "above" LUB$\{(a*b), (c*d)\}$

5. Draw the Karnaugh map for the 5-variable function:
{ 0,1,2,8,10,15,16,17,18,20,21,22,24,25,26,28,29,30
Draw in the cubes.

25



-6

Give the minimal representation of this function.

$a'd' + ae'' + c'e' + a'bcde + b'c'd' + c'e'$

$c'e' + ad' + ae' + b'c'd' + a'bcde$

(one or two on final)
(circuit diagrams)

6. Prove that if both $f: X \to Y$ and $g: Y \to Z$ are onto, then $g \circ f$ is onto.

20

-4

H_1: $(\forall y)(\exists x)\langle x,y\rangle \in f$

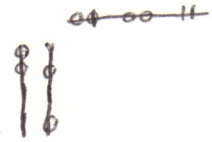
H_2: $(\forall z)(\exists y)\langle y,z\rangle \in g$

C: $(\forall z)(\exists x)\langle x,z\rangle \in g \circ f$

remove the $\forall$ first

1)	$(\forall z)[(\exists y)\langle y,z\rangle \in g]$	P
2)	$(\forall z)\langle b,z\rangle \in g$	~~ES~~ (1)
3)	$\langle b,c\rangle \in g$	US (2)
4)	$(\forall y)(\exists x)\langle x,y\rangle \in f$	P
5)	$(\forall y)\langle a,y\rangle \in f$	ES (4)
6)	$\langle a,b\rangle \in f$	US (5)
7)	$\langle a,b\rangle \in f \wedge \langle b,c\rangle \in g$	Ig (3,6)
8)	$\langle a,c\rangle \in g \circ f$	def^{n} of $\circ$
9)	$(\forall z)\langle a,z\rangle \in g \circ f$	UG (8)
10)	$(\forall z)(\exists x)\langle x,z\rangle \in g \circ f$	EG (9)

QED



1. Use mathematical induction to show

$P(n)$: $\frac{n^2(n+1)^2}{4} = 0^3 + 1^3 + 2^3 + 3^3 + \ldots + n^3$

20

$P(0)$: $0^3 = \frac{0^2(0+1)^2}{4}$ T

$P(m)$: $0^3 + 1^3 + \ldots + m^3 = \frac{m^2(m+1)^2}{4}$ — assumed

-3

$0^3 + 1^3 + \ldots + m^3 + (m+1)^3 = \frac{m^2(m+1)^2}{4} + (m+1)^3$ — add $(m+1)^3$ to both sides

$= \frac{m^2(m+1)^2}{4} + \frac{4(m+1)^3}{4}$ — multiply by $\frac{4}{4} = 1$

$= \frac{m^2(m+1)^2 + 4(m+1)^3}{4}$

$= \frac{m^2(m^2+2m+1) + 4(m+1)^3}{4}$

$= \frac{m^4 + 2m^3 + m^2 + 4(m+1)^3}{4}$

$= \frac{m^4 + 2m^3 + m^2 + 4(m^3 + m^2 + 2m^2 + 2m + m + 1)}{4}$

$= \frac{m^4 + 2m^3 + m^2 + 4m^3 + 4m^2 + 8m^2 + 8m + 4m + 4}{4}$

1) show $P(k)$
2) assume $P(m)$
3) derive $P(m+1)$
4) show $(\forall z)\, P(z) \to P(z+1)$ using CP

<over>

2. Prove that in any Boolean Algebra, $(a*b)+(a*b')=a$

6

$a = a*(b+b')$; C-1 and B-3

$a = (a*b)+(a*b')$; D-1

$(a*b)+(a*b') = a$; def$^{\underline{n}}$ of =

3. Prove that in any lattice, if $c \geq a$, then $c \geq a*b$

5

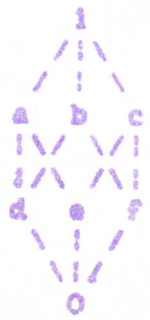
$\underline{a}$ is "below" $\underline{c}$, $\underline{a}$ is "above" $\underline{a*b}$ by def$^{\underline{n}}$ of $*$.

therefore $a*b \leq a \leq c$, and $a*b \leq c$ by transitivity. or equivalently $c \geq a$

4. If $f(x,y,z) = (x + y')*z'$, use the diagram at the right to compute: $f(a,d,c) =$ d

6

$f(d,a,0) =$ ~~1~~



-3

3) $c \geq a$
$a \geq a*b$ def$^{\underline{n}}$ $*$
$c \geq a*b$ transitivity

? $b*1 = b$

$$= \frac{m^4 + 6m^3 + 13m^2 + 12m + 4}{4}$$

$$= \frac{(m+1)^2(m+2)^2}{4}$$ — by long division!

which is $P(m+1)$, so $P(m) \to P(m+1)$ ... $\forall m$ ... and by mathematical induction $P(n)$ works for all $n > 0$, $\mathbb{N}$.

$$\frac{m^2(m+1)^2}{4} + \frac{4(m+1)^3}{4} = \frac{m^2(m+1)^2}{4} + \frac{4(m+1)(m+1)^2}{4}$$

$$= \frac{\cancel{4m^2(m+1)}\,(m^2 + 4m + 4)(m+1)^2}{4} = \frac{(m+1)^2(m+2)^2}{4}$$